

Policy Brief

Anticipating AI's Impact on the Cyber Offense- Defense Balance



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Executive Summary

The cyber domain touches nearly all systems and aspects of society, so any changes to the relative offense-defense balance in cyber could be very impactful. As a digital technology, AI can be expected to have a more direct effect on those balances than in other domains.

To assess how AI may affect the offense-defense balance within cyber, we collected arguments for an offensive or defensive bias in various aspects of cyber operations as well as arguments for what gives cyber its unique character. We then considered how varying levels of AI advancement might strengthen, weaken, or alter those arguments. The results of that analysis are grouped into five categories: Changes to the Digital Ecosystem, Hardening Digital Environments, Tactical Aspects of Digital Engagements, Incentives and Opportunities, and Strategic Effects on Conflict and Crisis.

There is no single answer to the question of whether AI will make cyber offense or defense dominant. Cyber attackers and defenders have too many different goals that can be achieved in multiple ways, but AI is likely to change the cyber landscape in ways that can be predicted and perhaps controlled to some extent.

Although AI will increase the scope of defensive tasks by making the digital ecosystem larger and more complex, it may also reduce the scope of defensive tasks in other ways, such as by decreasing the number of network connections to monitor. AI systems could replace known human weaknesses, but AI components are often vulnerable. AI components could also aggregate too much information or control into high-risk digital targets, and eliminating manual controls could reduce resilience during attacks. As system designers, acquisition officials, and users incorporate or implement AI, they will decide how much risk to accept along each of these lines.

AI also promises to further harden digital environments by performing tasks that currently overwhelm defenders. If these tasks can be done reliably by AI and if defenders can keep up with faster discoveries of new vulnerabilities and attack tactics, then defenders can take advantage of their ability to impose delays and frictions to gain more from AI than attackers. Doing so could prevent AI from enticing new threat actors and could limit the strategic benefits that aggressors might see from AI's increase in speed and scale. But that defensive advantage is far from guaranteed and there are several missteps that could push the balance toward offense instead of defense in the years to come.

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Introduction

Cyberattacks are already estimated to cause trillions of dollars in losses and damages per year, in addition to the hundreds of billions spent on cyber defenses.¹ And they've repeatedly brought down critical infrastructure, such as the energy grid in Ukraine as well as the Colonial Pipeline and hospitals across the United States.² But with digital technology so deeply integrated into all aspects of society, the potential impacts in the future could make these seem insignificant by comparison.

As a digital domain, cyber may be among the earliest and most transformed by AI but it is unclear what further changes will be. Will AI help defenders protect sensitive information, sustain operations, prevent threats and coercion, and harden critical infrastructure or will it enable and encourage attackers to perform even more devastating attacks?

To start answering these questions, we considered AI's impact on the factors that give advantages to cyber defenders and those that give advantages to cyber offensives. We collected arguments for the relative advantages of cyber attackers and defenders from across the literature. Some examples are that defenders determine the digital terrain, and that attackers choose who to strike. We also considered arguments for what gives cyber its character.³ Examples of those characteristics are that cyber tactics are fast, cyberspace has fuzzy borders, and strategic success is tied to audacity and initiative. We did not make judgments about the validity of any of the posed arguments; we simply collected them and considered how various levels of AI advancement could strengthen, weaken, or alter the arguments.

The complete list of arguments and the individual analysis of each one is available in a separate report.⁴ In this report, we outline the main ways that AI might impact cybersecurity from those analyses. We grouped those impacts into five categories described in the following sections: Changes to the Digital Ecosystem, Hardening Digital Environments, Tactical Aspects of Digital Engagements, Incentives and Opportunities, and Strategic Effects on Conflict and Crisis.

There is no obvious answer to the question of whether AI will be more successful in improving cyber offenses or cyber defenses. That question itself is too broad. For example, if AI were to drastically drive down the costs of all cyber activities, would that mean that it benefits defenders because defenders currently spend so much more? Would that still be a benefit to defenders if the attackers were able to use those lowered costs to conduct more attacks? Would that still favor attackers if those more

numerous attacks led to more data losses but fewer, or shorter-lived, critical system compromises? We may not be able to give a single definitive answer to such a broad question, but by considering AI's effect on the individual aspects of cyber we aim to clarify how AI could alter both offense and defense.

Changes to the Digital Ecosystem

AI is likely to reshape and rescale the defensive challenge. It will almost certainly increase the amount of code, services, and systems to protect, and it may simultaneously increase the complexity of those codebases, services, and systems. That could increase the interconnectedness of digital systems in ways that exacerbate the types of systemic vulnerabilities that already exist from complex and interconnected digital ecosystems.⁵ And if AI systems further centralize technology development to a few AI providers, that could also lead to more of the types of failures that repeat systematically across developers and the systems they develop.⁶

While scale, complexity, and aspects of interconnectedness increase the challenge for defenders, networks may reshape in ways that could be defensively advantageous. Defenders today struggle to identify illegitimate attacks among the many legitimate connections and commands from around the world. If AI providers become a central interface for many tasks and information exchanges, that could reduce the scope of network defense. For some systems, embedded AI could eliminate the need for connectivity altogether. A piece of critical infrastructure may no longer need remote access over the internet if it can operate independently, and smartphones may connect to fewer external sites to retrieve information or perform tasks if their internal models are sufficiently capable.⁷

These competing trends will reshape cybersecurity rather than make it more defensive or offensive. Defense has a chance to create a safer digital ecosystem as long as critical components manage to remain secure despite the pressures to implement AI models and agents that are notoriously vulnerable and fallible.⁸ Defenders need to reap the rewards of eliminating human weak links without creating scenarios where humans cannot intervene in a crisis. And defenders need to avoid creating AI models and systems that compile too much information or capability into targets that are easy to steal or easy to manipulate.

Hardening Digital Environments

If AI accelerates vulnerability discovery, it could overwhelm defenders—especially if those vulnerabilities are difficult to patch, such as hardware vulnerabilities.⁹ It is also possible that AI could do a worse job at implementing defenses but be implemented anyway for its low cost or speed.¹⁰

But overall, there are many opportunities for AI to help harden defenses. Cyber defenders are already overwhelmed with tasks both inside and outside their networks, and AI promises to help address this shortfall.¹¹ It will help clean and refactor an uncountably large number of lines of code.¹² It will help defenders adhere to volumes of standards and best practices. It will help implement patches and updates faster. It will help defenders understand and assess their networks and the devices on them, anticipate the types of risks on those networks and systems, and help perform assessments with more realistic AI-enabled red teams.

As long as vulnerability discovery does not become faster and more novel than AI-enabled defenders are able to keep pace with, AI is likely to harden digital environments.

Tactical Aspects of Digital Engagements

It is important to keep in mind that cyberattacks already occur at internet scales and at incredible speeds. What AI threatens to provide is more independence and autonomy among those scaled attacks. Still, AI could accelerate some of the more time-consuming aspects of offensive techniques and tactics, such as reverse engineering, aspects of reconnaissance, and exploit development.¹³

Defense can also benefit from accelerated reverse engineering, patch implementation, and script writing. Further, defenders can design their systems to limit the speed and scale of various tasks or procedures; they can throttle suspicious connections, or require human confirmations before allowing dangerous activities such as deleting or encrypting critical files. Those defensive systems can also use large servers to run powerful AI agents, whereas attackers may be restricted to smaller local computations or limited data transfers to remote servers to stay below alert thresholds. And finally, layered defenses appear promising for repelling large numbers of intelligent attackers.¹⁴

A primary challenge for defenders is ensuring that any actions they take are reliable. An unreliable patch or configuration change could end up inflicting the damage they seek to prevent.¹⁵ If AI can help design or test defenses to ensure reliability, then defenders have much to gain from AI at a tactical level. In either case, defenders have several advantages that can be used to limit attackers' ability to use AI effectively.

Incentives and Opportunities

AI will provide new threat actors — including cyberterrorists, individuals, and rogue nations — with enhanced attack tools that are even easier to use than the ones that are widely available today. If AI does not adequately harden defenses, then those additional threat actors could become more difficult to restrain and more numerous than the advanced threat actors today. AI could also remove some of the signals that are useful for attribution, such as aspects of how the code is written, languages used, or hours when attackers are working.¹⁶

AI may also enhance some aspects of attribution for defenders by drawing from disparate and complex data, but attribution will probably continue to be based largely on signals such as intercepted communications, ownership of infrastructure, and motive.¹⁷

AI will probably make cyberattacks easier to attempt for a larger number of less skilled threat actors while making attribution more difficult. Whether those attackers will find greater payoff from their efforts depends largely on whether AI manages to harden defenses to such a degree that the new AI attack tools are less effective than the attack tools that exist today.

Strategic Effects on Conflict and Crisis

Strategic aspects of cyber are typically slower than tactical ones, but they may be difficult for AI to accelerate or improve. Humans will probably want to remain in control of strategic decisions. Even an incredibly capable AI may not be able to resolve disagreements about incompatible values and priorities among the many stakeholders who contribute to an organization's defense or decisions about attack.

Aggressors may choose to attack more often if cyberattacks can be prepared and conducted more quickly.¹⁸ Rapid development may make tailored cyberattack options more available to pair with evolving conflicts or crises. And the ability to attempt more cyberattacks may make them more viable overall. If the attacker only needs any

disruption to occur, then AI may help to attempt many disruptions. That may also increase the odds of accidental damages or escalation.¹⁹

Defenders may be able to inspect the AI attack agents to learn not only their capability and signatures but also their internal decision processes. That could reduce miscommunications or miscalculations and provide defenders with some advantages. But that testing and analysis may take time, and it will rarely be definitive because attackers will likely be able to update the attack agents' decision processes throughout a conflict or crisis.

It appears that there are strategic advantages that will accrue to attackers. Aggressors will likely be able to conduct attacks on more amenable timelines with more attempts while more rapidly recovering lost access and replacing burned capabilities. The prospect of more accidental damages and escalations would also not be welcomed among defenders.

Recommendations

There are many ways that AI could push the offense-defense balance of cybersecurity for better or worse, but none are a foregone conclusion. Defenders, developers, and consumers can take many actions now to avoid the worst outcomes and enable the best.

Incentivize Reliability Over Originality

Defenders have many tasks that are not individually difficult but that need to be performed reliably. They need to revise vast codebases, implement volumes of standards and best practices, adjust countless configurations, and inspect numerous alerts. Errors in implementing these defensive tasks can be as damaging as the attacks they seek to prevent. Attackers typically care less about errors but rely more on novelty and originality. AI developers today are rewarded for demonstrating novel capability more than they are for reliability and consistency. For defense to benefit more than offense, AI developers and researchers need to prioritize reliability over novelty. They should be rewarded for performing simple tasks like adjusting configurations, writing patches, or summarizing suspicious events or code without hallucinations or errors. That will require changes in funding allocation, benchmark development, and news coverage.²⁰

Fund Provable Security

The scenario where the balance shifts most toward offense is where AI becomes adept at discovering new vulnerabilities or new attack tactics. That scenario becomes much less worrisome if substantial portions of the digital environment are made provably secure. In that case, even very advanced AI systems would not be able to weaken certain aspects of their targets. Defenders could build some secure barriers and focus their efforts on fewer potentially vulnerable systems and components.

Currently, only a few small pieces of software and hardware are provably secure, and encryption algorithms are believed to be difficult to break but are not provably difficult to break.²¹ To protect against the worst possible future scenarios, funding agencies should invest more in formal methods, exhaustive testing, designing systems to take advantage of formally-verified components, and perhaps simplifying some aspects of critical systems.

Fund Live Patching

Another worrisome scenario is where AI provides capability to both offense and defense but the defense struggles to implement those advances due to practical or organizational limitations. For example, defenders already struggle to implement patches and updates quickly because of the downtime and restarts that those updates can require.²² Faster patches and updates would put even more stress on already taxed system administrators. Patching systems without downtime is possible and will need to be further prioritized and funded if the patch rate needs to increase significantly.²³

Establish and Maintain Standards for Security and Reliability of AI Systems

There are intense pressures to implement AI systems and components in nearly all sectors and industries to reduce costs, gain technological advantage, supplement expertise, or accelerate operations. But AI systems and components can also be vulnerable and lucrative targets. Components for critical systems typically require far more reliability and security than AI components have demonstrated.²⁴ AI's may be used in those systems to replace error-prone humans rather than reliable electrical or mechanical components; however, new standards will need to be developed to ensure that the resulting systems maintain the same levels of security and reliability.²⁵

Maintain the Option for Human Control

While humans are both error-prone and common weak links in security, they are also adaptable and resilient in a crisis. That resilience is a critical aspect of defense that limits the damage from cyberattacks. Acquisition officials and system designers should require that their systems maintain manual modes of operation in order to limit the damage from potential digital attacks and failures. The operators of those systems should also maintain the knowledge and skills to operate those systems in the event of a cyberattack or digital failure.

Assess Compilation and Aggregation Risks

AI systems can be most capable when they are given the maximum amount of information or the maximum amount of control. For example, an AI system that has read all of the classified reporting in an organization or that monitors all of a user's activity would be the most well-informed. Likewise, an AI system that has access to all the controls of an industrial control system would be most capable, but those would also be unprecedentedly risky practices and lucrative targets. System designers and acquisition officials should not create unprecedented risks without commensurate value and security.

Enable Air-Gapping and Reduced Connectivity

Defenders may benefit from AI's ability to reduce the number and diversity of external resources and connections that systems and users require. System designers will need to recognize the defensive advantage of limiting connectivity and use the capacity of AI systems to reverse the trend toward interconnectedness.

Design Systems to Enhance Defensive Advantage

Digital systems have been designed with defensive slowdowns and interventions such as password lockouts or bandwidth throttling that limits bots from scraping data from websites. These types of interventions limit the advantage that offense can receive from AI without hindering the defense's ability, and there are many ways to introduce breakers, slowdowns, and human approvals into digital systems. Existing and emerging techniques should be included in standards and best practices that acquisition officials require of system designers. However, further research and funding is needed to advance new techniques.

Practice AI-Induced Cyber Emergencies

Some of the worst-case scenarios involve rapidly developing offensive capabilities. Defenders may not be able to immediately respond to a flood of risks if, for instance, an AI system suddenly finds many new vulnerabilities of a certain class or develops a set of new attack tactics. In the time before mitigations are developed, the owners and operators of critical systems could be vulnerable in ways and to degrees that they have not experienced and have not prepared for. To prepare for new threats, they need to conduct realistic table-top exercises to identify the lines of communication and decision-making required in an emergency. Those interventions could include temporarily disabling critical systems to limit access and damage. A real-world crisis is not the time for first considering the decisions and actions necessary to respond.

Conclusion

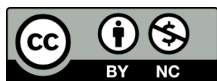
There is no single clear answer to whether AI will advantage cyber offense or defense, but there are opportunities for consumers and technology developers to shift the balance toward defense. AI is likely to harden the digital ecosystem and individual digital environments as long as developers resist the urge to shortcut security for the sake of increased convenience and reduced costs. Defenders may look to AI to help implement patches and configuration changes, especially if they face floods of new AI-discovered vulnerabilities and offensive tactics. At a more tactical level, defenders can design their systems to maximize the defensive benefits of AI while minimizing its benefits for offense. As long as AI is reliable enough to harden digital systems at scale, new threat actors may not find AI attack tools as valuable as they hope, despite the strategic benefits that faster attack timelines and greater numbers of attackers provide.

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